```
D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-PIKAN_Official\experiments\Volterra1D_MCPI.py
The network model is cPIKAN
Layers: ModuleList(
  (0): ChebyKANLayer(in_features=1, out_features=10, degree=3)
  (1): ChebyKANLayer(in_features=10, out_features=10, degree=3)
  (2): ChebyKANLayer(in_features=10, out_features=1, degree=3)
)
Total trainable parameters: 480
Model on device: cuda:0
Start training...
Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
Training: 100%|██████████| 2000/2000 [00:08<00:00, 224.41epoch/s, Loss=5.37e-06, LR=1.0e-03, Best=5.37e-06]
Training finished. Best loss: 4.26e-06. Model saved to results/best_model.pth
Loss history saved to results/loss_history.npy
Training time: 9.7172372341156s
Relative loss: 0.002688765525817871
L2 loss: 0.0017108652973547578
Parameter containing:
tensor([[[ 0.0745,  0.1399,  0.4815, -0.0701],
         [-0.0268, -0.5645, -0.2220,  0.1873],
         [-0.0728,  0.3010,  0.3437,  0.0316],
         [-0.0809, -0.2038, -0.0675,  0.7762],
         [ 0.1887, -0.1553, -0.5296, -0.0871],
         [-0.6213,  0.1551,  0.3519, -0.0735],
         [ 0.2834, -0.0101, -0.2481, -0.2300],
         [ 0.0050,  0.3313, -0.2814, -0.3142],
         [ 0.6983, -0.1931, -0.6662, -0.4118],
         [ 0.3017, -0.3456, -0.5570, -0.0485]]], device='cuda:0',
       requires_grad=True)
Parameter containing:
tensor([[[-6.7840e-03,  1.9651e-01,  1.4338e-02, -1.7278e-01],
         [ 5.1527e-02, -2.1824e-01, -6.8432e-02,  2.3374e-01],
         [ 2.1518e-02,  1.0207e-01, -1.8296e-02, -7.5650e-02],
         [ 1.3214e-02, -2.1782e-01, -2.8832e-02,  1.9952e-01],
         [-2.8221e-02,  1.9321e-01,  4.8931e-02, -1.3227e-01],
         [ 3.8729e-02, -1.7609e-01, -3.3941e-02,  1.7696e-01],
         [ 1.5235e-02, -1.0754e-01, -1.6835e-02,  1.1033e-01],
         [ 5.2406e-02, -2.2113e-01, -1.4472e-02,  1.5872e-01],
         [-2.3808e-02,  1.5580e-01, -1.5093e-02, -2.0617e-01],
         [-2.9784e-02,  2.2708e-01, -7.5029e-03, -1.5963e-01]],

        [[-4.3594e-02, -2.0905e-01,  5.1447e-02,  1.7847e-01],
         [ 3.6075e-02,  2.1980e-01, -8.3238e-03, -1.6993e-01],
         [ 3.6490e-02, -2.1742e-01, -1.0001e-02,  2.3679e-01],
         [ 4.8752e-02,  2.1604e-01, -4.9631e-02, -2.3948e-01],
         [-2.2440e-02, -2.1445e-01,  4.1611e-02,  1.7969e-01],
         [-1.6528e-02,  1.6668e-01, -6.1345e-02, -1.7084e-01],
         [ 4.1345e-02,  1.7826e-01,  3.6236e-03, -1.0930e-01],
         [-1.7198e-02,  2.2572e-01, -2.3732e-02, -2.1532e-01],
         [-5.9452e-03, -2.5991e-01, -3.9857e-03,  2.0562e-01],
         [-6.1364e-02, -2.1292e-01,  6.0628e-02,  1.8801e-01]],
```

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        [[-2.1106e-02,  1.4459e-01, -2.9152e-02, -1.6395e-01],
         [ 8.7111e-02, -2.8640e-01, -6.3167e-02,  2.7409e-01],
         [ 1.9738e-03,  7.5111e-02,  5.9232e-03, -2.1289e-02],
         [ 2.3566e-02, -2.2753e-01,  4.7785e-03,  2.0334e-01],
         [-8.4292e-02,  2.3454e-01,  6.0908e-02, -1.8816e-01],
         [ 5.1548e-02, -1.3395e-01, -3.1465e-02,  1.9251e-01],
         [ 2.0629e-02, -1.4550e-01, -1.3567e-02,  1.0799e-01],
         [ 3.6495e-02, -2.0889e-01, -3.8405e-02,  1.5008e-01],
         [-4.6961e-02,  1.8979e-01, -3.7319e-02, -1.8606e-01],
         [-7.5932e-03,  1.7888e-01,  1.2197e-02, -1.9275e-01]],

        [[-4.7173e-02, -6.8566e-02, -2.9747e-02, -9.1193e-03],
         [ 9.7188e-02, -5.9136e-02,  1.7256e-02, -8.2888e-02],
         [ 8.4958e-02, -1.5736e-01,  2.9444e-02,  1.2582e-01],
         [-1.7724e-03,  1.0651e-01,  9.5512e-03, -8.2812e-02],
         [-4.4736e-02, -1.2232e-04,  4.9421e-03,  2.3036e-02],
         [ 3.5959e-02, -1.3893e-02, -2.4958e-02, -2.1494e-02],
         [-7.3780e-03, -1.1574e-02, -3.0701e-02, -7.6595e-03],
         [ 2.7127e-02,  1.4098e-01, -6.0487e-02, -1.7915e-01],
         [-2.0010e-02, -7.5765e-03,  4.3555e-02,  1.3827e-02],
         [-1.3299e-02, -4.2309e-02,  6.1436e-02,  4.5600e-02]],

        [[-1.9202e-02, -6.7173e-02, -4.9907e-03,  3.9450e-02],
         [-3.3929e-03,  1.3803e-01, -2.4862e-02, -1.3081e-01],
         [ 3.8485e-03, -8.8284e-03, -3.2663e-02, -4.0540e-02],
         [ 8.8323e-03,  1.6883e-02, -8.7268e-03, -4.1044e-02],
         [-3.8565e-02, -5.9242e-02,  3.0003e-03,  7.9491e-02],
         [ 7.5283e-03,  1.1813e-01, -2.1750e-02, -2.9667e-02],
         [ 6.6171e-03,  2.6665e-02, -1.6049e-02, -8.3942e-02],
         [ 8.8707e-03,  8.4235e-02, -4.2020e-02, -6.2570e-02],
         [ 1.9757e-03, -7.6110e-02, -1.0439e-02, -6.8920e-03],
         [-2.8332e-02, -6.7515e-02, -1.6569e-02,  8.1945e-02]],

        [[-3.1643e-02,  4.3860e-02, -2.7217e-02, -2.0330e-02],
         [ 4.8468e-02, -2.9329e-02,  6.4977e-02,  6.8084e-02],
         [ 4.4111e-03, -1.3318e-02, -6.9204e-02,  9.0553e-03],
         [ 1.4735e-02, -8.6248e-02,  8.1417e-02, -2.2368e-02],
         [-2.3985e-02,  1.5805e-02, -3.1965e-02, -6.4261e-02],
         [ 4.9975e-02, -2.2070e-03,  6.1830e-02,  3.6097e-02],
         [ 2.3105e-02, -2.8917e-02,  1.8024e-02, -1.4798e-02],
         [ 2.1275e-02, -5.7805e-02,  1.1677e-01, -2.3831e-02],
         [-6.2193e-03,  1.7946e-02, -1.3497e-02, -1.8333e-02],
         [-4.8508e-02,  9.7169e-03, -5.0186e-02, -1.7564e-02]],

        [[ 3.0183e-02, -1.1551e-02,  6.9241e-02,  3.7637e-02],
         [ 3.5938e-02,  5.5294e-02,  1.8548e-02, -1.0247e-01],
         [-6.0642e-02,  3.7195e-02,  2.6653e-02,  4.8174e-03],
         [-1.7401e-02,  1.4268e-02, -2.9572e-02, -1.6862e-02],
         [-1.0137e-02, -2.9666e-02,  4.3988e-03, -1.1429e-02],
         [ 7.1776e-03,  2.8408e-02,  1.7978e-03, -2.5004e-02],
         [ 5.2224e-02,  3.1673e-02,  1.4327e-02, -4.3604e-02],
         [ 4.6239e-02, -1.8891e-03, -7.8604e-02, -4.3118e-02],
```

```
          [ 5.5103e-03, -7.8386e-03,  4.9082e-02, -1.4404e-03],
          [-6.5871e-02, -3.5417e-02,  2.0727e-02,  1.1159e-02]],

         [[-9.7937e-02, -2.5817e-02,  2.1664e-02,  4.8403e-02],
          [ 5.3460e-02,  1.0146e-01, -1.3729e-02, -2.7395e-02],
          [-1.1121e-02,  2.5295e-02,  4.2714e-02, -7.6282e-02],
          [ 7.4753e-02, -3.8129e-02, -6.4887e-02,  1.5177e-02],
          [-4.4118e-02,  4.9875e-02,  5.1829e-02,  5.9227e-02],
          [ 2.3126e-02,  2.9540e-02, -1.1425e-02, -1.6971e-02],
          [-1.5722e-02,  5.0479e-02, -1.2797e-02, -1.0758e-02],
          [ 4.3953e-02, -3.0287e-02, -5.3185e-02,  2.7406e-02],
          [-5.1317e-02,  1.7096e-03,  3.9987e-02,  1.6508e-02],
          [-5.0405e-02, -3.9224e-02,  7.0049e-03, -1.8139e-03]],

         [[ 1.8854e-02, -3.1005e-02, -1.4346e-01, -1.3111e-02],
          [ 4.8128e-02,  9.5188e-02,  3.1888e-01, -6.3099e-03],
          [-3.1902e-03,  5.6501e-02,  8.3029e-02,  1.4606e-02],
          [-2.7941e-03,  1.0802e-02,  9.9911e-02, -5.2735e-02],
          [-2.9715e-02, -4.8516e-02, -2.2424e-01,  2.9492e-03],
          [-1.3875e-02,  3.2825e-02,  1.5896e-01, -2.8451e-02],
          [ 4.2830e-02, -4.6958e-03,  2.0588e-01, -4.7125e-03],
          [ 4.3411e-02,  3.4655e-02,  7.5844e-02, -9.1669e-02],
          [ 2.9520e-02, -1.7424e-02, -1.0813e-01,  3.4253e-02],
          [-2.2433e-02, -5.4554e-02, -1.5848e-01,  5.5763e-02]],

         [[-9.7766e-03, -2.7749e-02,  4.2758e-02,  7.0078e-02],
          [ 3.7113e-02,  1.1871e-01,  1.0754e-02, -1.3787e-01],
          [ 3.0425e-02,  1.6949e-02, -4.5099e-02, -2.1033e-02],
          [ 1.7627e-02,  7.3789e-02,  5.4698e-02, -3.9911e-02],
          [-2.5867e-02, -5.3005e-02, -1.5786e-02,  5.2214e-02],
          [ 4.2168e-02,  5.6112e-02,  2.2216e-03, -6.3565e-02],
          [ 4.8667e-02,  3.6768e-02, -1.4130e-02, -2.9176e-02],
          [ 3.6287e-02,  9.1530e-02,  1.5888e-01, -2.6286e-02],
          [-8.7252e-02, -5.3163e-02, -4.7829e-02,  3.4788e-02],
          [-2.0315e-02, -9.0121e-02, -2.2883e-02,  4.3727e-02]]],
       device='cuda:0', requires_grad=True)
Parameter containing:
tensor([[[-0.0041, -0.0191, -0.0288,  0.1333]],

        [[ 0.0048,  0.0268, -0.0277, -0.0718]],

        [[ 0.0342,  0.0677,  0.0204, -0.1068]],

        [[ 0.0663,  0.0060, -0.0822, -0.0775]],

        [[-0.0144, -0.0050, -0.0003,  0.0669]],

        [[ 0.0354,  0.0179, -0.0221, -0.0973]],

        [[ 0.0079,  0.0070,  0.0206, -0.0557]],

        [[ 0.0090, -0.0234, -0.0493, -0.0953]],

```

```
          [[ 0.0553,  0.0039, -0.0499,  0.0466]],

          [[-0.0262, -0.0234, -0.0184,  0.0521]]], device='cuda:0',
         requires_grad=True)
====================================================================================
======
Layer (type:depth-idx)                   Output Shape              Param #
====================================================================================
======
CPIKANModel                              [1, 1]                    --
├─ModuleList: 1-1                          --                        --
│    └─0.weight                                                        ├─40
│    └─1.weight                                                        ├─400
│    └─2.weight                                                        └─40
│    └─ChebyKANLayer: 2-1                   [1, 10]                   40
│    │    └─weight                                                      └─40
│    └─ChebyKANLayer: 2-2                   [1, 10]                   400
│    │    └─weight                                                      └─400
│    └─ChebyKANLayer: 2-3                   [1, 1]                    40
│    │    └─weight                                                      └─40
====================================================================================
======
Total params: 480
Trainable params: 480
Non-trainable params: 0
Total mult-adds (Units.MEGABYTES): 0.00
====================================================================================
======
Input size (MB): 0.00
Forward/backward pass size (MB): 0.00
Params size (MB): 0.00
Estimated Total Size (MB): 0.00
====================================================================================
======

进程已结束，退出代码为 0

```